SIMATS ENGINEERING

ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I G. Amrutha (192511250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - Analyze deployment models, business benefits, challenges, and practical examples.
7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
   - Discuss AI/ML cloud services, architecture, and industry applications.

CLOUD STORAGE TECHNOLOGIES: OBJECT, BLOCK, AND FILE STORAGE COMPARED.

1) Introduction:

Cloud Storage is a technology that allows individuals and organizations to store, manage, and access data through internet-based computing infrastructure. Instead of keeping data only on local hard drives or physical servers, Cloud Storage uses remote data centers operated by cloud service providers.

2. objective

the main objectives of this report are:

=) To understand the concept of cloud storage technologies.

=) To explain object, Block, and File storage.

=) To compare their architecture, performance, scalability, and use cases.

=) To understand how different industries use these storage technologies.

=) To identify the advantages and limitations of each storage type.

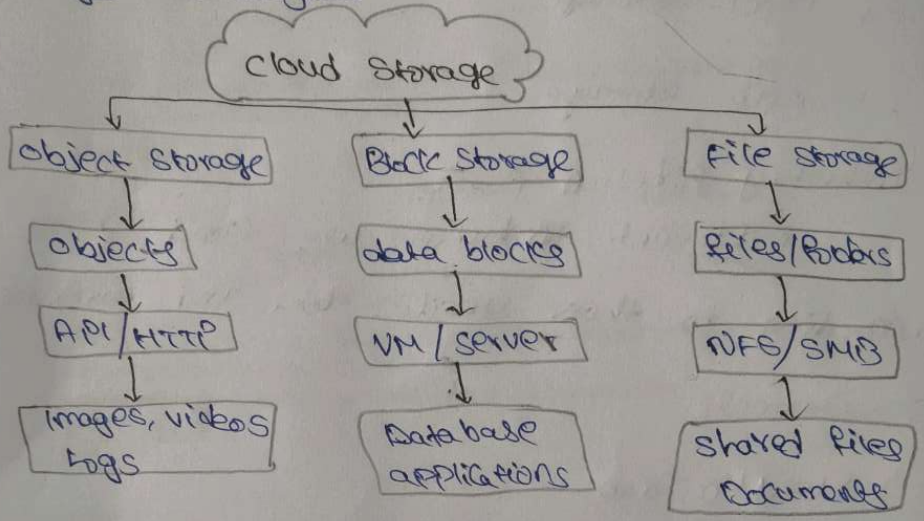
=) To understand how organizations can select the appropriate storage solution for their requirements.

3. Comparison of object, Block, and File storage

Feature	object storage	Block storage	File storage
Data organization	objects	Blocks	Files and directories
Access method	APIs / HTTP	Storage volumes	File protocols
Scalability	very high	High	High
Typical data	Images, videos backups	Database data	Documents.
Example workload	Data archive.	Database	Shared office files.

4. Architecture or Process Diagram.

The basic architecture of the three storage technologies can be represented as:



Object Storage:

Applications → API Request → Object Storage → object retrieved.

Block Storage:

Application → operating system → Storage Volume → Data blocks

File Storage:

User/Application → File Protocol → File System → File/Folder

Real-world use cases

=) Object Storage

=) Block Storage

=) File Storage

Industry Adoption Patterns

Different industries use Cloud Storage according to their specific data and performance requirements.

=) Health Care

=) media and Entertainment

=) Education

=) Retail and E-Commerce

=) Software and Technology

Advantages and Limitations.

object Storage

Advantages:

* Excellent scalability.
* Suitable for very large datasets.
* Good for unstructured data.
* Supports extensive metadata
* useful for backup and archival

Limitations:

* not ideal for applications requiring frequent data.
* usually accessed through APIs rather than file system
* performance characteristics differ from high-performance block storage.

Block Storage

Advantages:

* High performance
* Low latency.
* Suitable for databases
* Works well with virtual machines.

Limitations:

* Can be more expensive for large-scale data storage.
* Requires more management by the operating system.
* Not naturally designed for direct file sharing.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism